

AI Real Estate Recommendation Agent

Simple Problem Formulation and Environment Classification

1. Problem Formulation

Element	Description
Initial State	The agent receives buyer requirements such as budget, preferred location, property type, number of bedrooms, and other preferences. It has no prior assumption about which properties are most suitable.
Actions	Collect and organize buyer requirements; search property listings; filter properties by budget, location, and property type; compare properties; rank suitable properties; generate recommendations; and update recommendations using buyer feedback.
Transition Model	When the agent analyzes new property listings or receives buyer feedback, its knowledge of suitable properties, rankings, and recommendations is updated.
Path Cost	The cost may be measured by search time, number of properties reviewed, or effort required. The objective is to use the buyer's time efficiently.
Goal Test	The buyer receives one or more properties that closely match the stated budget, location, property type, bedrooms, and other requirements.
State Space	All possible combinations of buyer requirements, available property listings, filtered properties, ranked properties, recommendations, and updated recommendations.
Search Space	All possible sequences of collecting requirements, searching listings, filtering properties, comparing options, ranking results, generating recommendations, receiving feedback, and updating the plan.
Agents	Multi-agent. Multiple buyers, sellers, real estate agents, and competing platforms can act within the same real estate environment.

2. Environmental Properties

Property	Classification	Why
Observability	Partially Observable	The agent only knows the information provided by the buyer and available property listings. It cannot see unstated preferences or all of the buyer's decision factors.
Determinism	Stochastic / Non-deterministic	Past information cannot guarantee the outcome. A property may be sold, its price may change, or the buyer may reject the recommendation.
Episodes	Sequential	Earlier searches and buyer feedback affect later recommendations, so decisions are connected rather than isolated.
Change	Dynamic	The real estate market keeps changing independently. New listings appear, prices change, and properties can be sold.
Values	Discrete & Continuous	Property types and some filters are discrete, while values such as price and area can be continuous.
Agents	Multi Agent	Multiple buyers, sellers, real estate agents, and competing platforms can act in the same market at the same time.

3. Choosing the State Type

Option	Suitability	Explanation
Multi-state	Most suitable	The agent moves through many possible states: buyer requirements, property listings, filtered properties, ranked properties, recommendations, and updated recommendations.

Final Answer: The AI Real Estate Recommendation Agent is best classified as a **Multi-state agent** because it passes through multiple changing states while processing buyer requirements, analyzing property listings, detecting suitable options, creating recommendations, and updating them using buyer feedback.